Stochastic Deep Supervision for Gallery-Consistent Early-Exit Face Recognition

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One ViT, trained once: exit at *any* block, match a gallery enrolled *once*, keep *one* threshold.

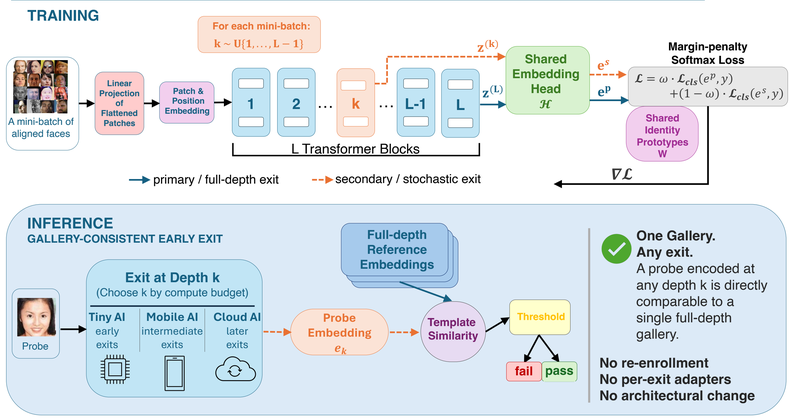
$11 \rightarrow$ **87**	$0 \rightarrow$ **93**	$\leq$ **0.4**	**1** τ	**+3-5%**
IJB-C TAR at quarter depth, ViT-S/12 (Baseline → SD-ViT)	quarter-depth probe against a gallery enrolled at full depth	IJB-C points from the conventionally trained model	one verification threshold, usable from quarter depth on	training time, no inference overhead

1 The problem

- One face recognition model must serve budgets from **edge devices to datacenters**, but a ViT always runs **all** L **blocks**.
- Early-exit networks give one model many budgets, yet they come from **closed-set** classification: every exit has its **own classifier**.

FR is **open-set**: a probe is matched against a **gallery enrolled once**. Early-exit FR therefore needs **gallery consistency**: every depth must embed into **one shared space**.

2 SD-ViT: one head for every depth

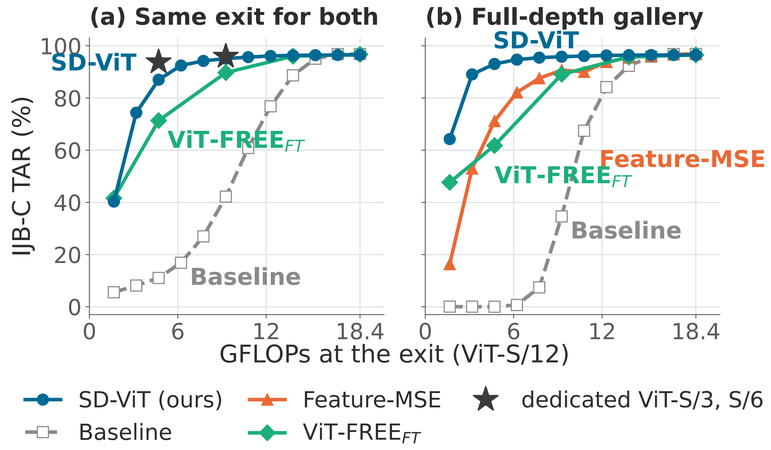


At every step, decode the full-depth tokens **and** one **uniformly sampled** block with the **same** head $\mathcal{H}$, and score both against the **same** prototypes W:

$$\mathcal{L} = \omega\,\mathcal{L}_{\mathrm{cls}}\big(\mathcal{H}(z^{(L)}), y\big) + (1-\omega)\,\mathcal{L}_{\mathrm{cls}}\big(\mathcal{H}(z^{(k)}), y\big), \quad k \sim \mathcal{U}\{1, \ldots, L-1\}$$

- **No teacher**, no per-exit heads, no adaptation layers.
- Both exits come from **one forward pass**.
- Inference: run k blocks, apply $\mathcal{H}$, **match the full-depth gallery**.

3 One network, every budget



IJB-C TAR at FAR 10^{-4}, ViT-S/12. The Baseline's early blocks **collapse**, and against a full-depth gallery (b) they are **unusable** below half depth. Dedicated ViT-S/3 and S/6 (★) win at their own budget, but N budgets then need N **networks and** N **galleries**.

IJB-C, Baseline → SD-ViT	$\frac{1}{4}$ depth	$\frac{1}{2}$ depth	full
ViT-S/12, WebFace4M	11.1 → **87.0**	42.2 → **95.1**	96.9 / 96.5
ViT-B/24, WebFace4M	21.6 → **91.8**	72.0 → **96.3**	97.0 / 96.7
ViT-B/24, WebFace12M	12.1 → **95.0**	55.9 → **97.1**	97.7 / 97.6

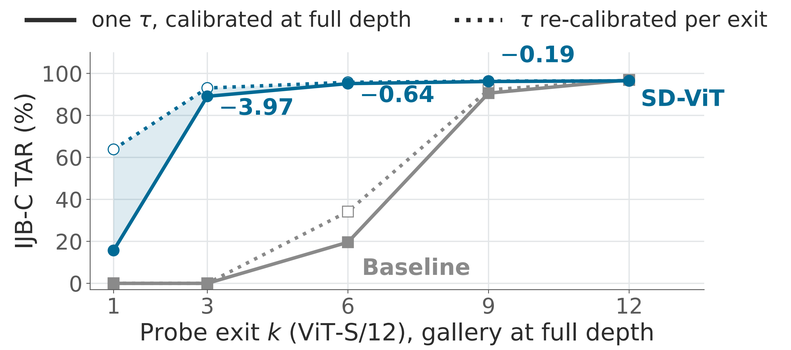
Scope

- Budget-selected exits; per-image policies are future work.
- Constant-width ViTs; also shown on KP-RPE.
- One seed per configuration.

Code & models

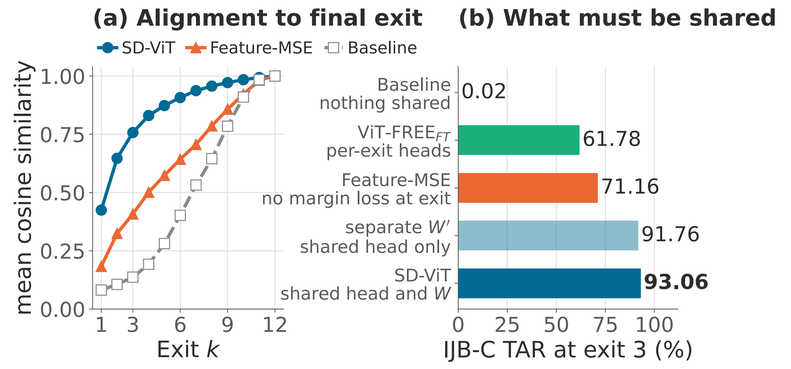
github.com/gurayozgur/SD-ViT

4 One gallery, one threshold



One τ=0.382, calibrated on **full-depth** development scores at FAR 10^{-4}, is applied **unchanged** at every exit. The realized FAR stays **below target** at every early exit. The Baseline is operable only from exit 9.

5 Why it works



(b) uses exit-3 probes against a full-depth gallery. The **shared head** places every depth in one space, the **margin loss** makes each depth discriminative in it. Supervising **all 11 exits** every step is on par, at 7.4× the extra training cost.